BTVNANOCOMMISSIONING · ZB_ZC_RATIO WORKFLOW

Z(→ℓℓ) + b/c AK4 jets in Run 3

Summer22 DY→ℓℓ MC with AK4 jets, 20 files per sample: samples, cutflow, kinematics, flavour and pileup checks

19 October 2026

Campaign and setup

Goal: measure σ(Z+b) / σ(Z+c) with an AK4 jet recoiling against Z→ee/μμ (switched from the boosted AK8 jet after the group meeting). First step: tune the selection on DY→ℓℓ MC.

MC CAMPAIGN

Summer22

Run3Summer22 = 2022 eras C–D (pre-EE)

FORMAT

NanoAODv12

130X MC conditions

ENERGY (√s)

13.6 TeV

Run 3 pp collisions

NORMALIZATION

7.98 fb^{-1}

2022 C–D golden JSON; brilcalc check pending

MC only, no data yet. Processed: 20 files per sample (the first 20 files of each sample in the fileset, 100 of 1887 DY files; run_zbzc_ak4_20files.sh). Code: BTVNanoCommissioning, branch coffea_machine, workflow Zb_Zc_ratio (AK4, commit 31ba6bc, with the GenJet-match counters).

MC samples: DY→ℓℓ, stitched in pT(ℓℓ)

```
/DYto2L-2Jets_MLL-50[_PTLL-X]_TuneCP5_13p6TeV_amcatnloFXFX-pythia8/
  Run3Summer22NanoAODv12-130X_mcRun3_2022_realistic_v5-vN/NANOAODSIM
```

Sample	LHE pT(ℓℓ) used	σ [pb]	DAS files	DAS events	20 files: events
Inclusive (MLL-50)	< 100 GeV	6346	525	93.8 M	3,266,113
PTLL-100	100–200 GeV	105.0	418	56.4 M	2,933,762
PTLL-200	200–400 GeV	11.24	335	44.7 M	2,735,083
PTLL-400	400–600 GeV	0.643	424	41.6 M	1,560,230
PTLL-600	> 600 GeV	0.0994	185	9.50 M	1,236,732

PTLL-X samples are inclusive above X, so each is kept to its own LHE pT(ℓℓ) slice. σ(PTLL-X) = 6346 pb × fraction of inclusive events with pT(ℓℓ) > X (measured on 27.7 M inclusive events). "20 files: events" = events of 20 files inside the slice. With no boosted-jet requirement the inclusive sample (pT(ℓℓ) < 100 GeV) gives 87% (Z→ee) / 89% (Z→μμ) of the scaled yield from only 5–6% of the raw selected events, so its large per-event weights limit the MC statistics.

Event selection, per channel

Z→ee / Z→μμ (unchanged)

- Trigger: HLT_Ele30_WPTight_Gsf / HLT_IsoMu24
- Leptons pT > 25 GeV, |η| < 2.4; leading pT > 35 GeV
- e: tight cut-based ID, IP cuts, EB–EE gap veto
- μ: tight ID, PF isolation < 0.15
- Opposite sign (OS)
- 71 < m$\ell\ell$ < 111 GeV
- PuppiMET < 50 GeV

Leading AK4 jet ("jet" step)

- AK4 PUPPI jets, ΔR > 0.4 from selected leptons
- Tight-lepton-veto jet ID (with NanoAODv12 jetId fix)
- pT > 30 GeV, |η| < 2.5; ≥ 1 such jet, leading one used
- No Z pT requirement

Weights and flavour

- genWeight × pileup × lepton ID/iso/reco SFs
- No trigger SF yet
- Jet flavour from hadronFlavour: b (5), c (4), light; no bb/cc split for AK4
- Also labelled by |partonFlavour|: b, c, uds, g, none (pages 21+, CMS recommendation on p. 21)
- Tagger: PNet AK4 B, CvL, CvB

Cutflow, Z→ee: raw MC events per step

Sample	all	trigger	leptons	OS	Zmass	MET	jet	jet / MET
Inclusive	3,266,113	541,950	165,143	164,108	158,226	150,078	39,242	26.1%
PTLL-100	2,933,762	840,163	238,941	236,499	218,939	193,805	186,102	96.0%
PTLL-200	2,735,083	929,177	344,674	339,940	309,699	256,244	252,727	98.6%
PTLL-400	1,560,230	576,282	264,078	259,407	233,151	171,946	171,561	99.8%
PTLL-600	1,236,732	306,140	138,357	135,652	112,766	73,911	73,866	99.9%

723,498 raw MC events pass the full Z→ee selection; the most (35%) come from PTLL-200. The AK4 jet step keeps 26% of inclusive (low-pT) Z events and ≥ 96% at pT($\ell\ell$) > 100 GeV.

Summer22 DY→$\ell\ell$ MC · 20 files per sample · AK4 jets · "all" = events in the sample's LHE pT($\ell\ell$) slice · 5

Cutflow, Z→μμ: raw MC events per step

Sample	all	trigger	leptons	OS	Zmass	MET	jet	jet / MET
Inclusive	3,266,113	690,771	290,730	290,720	280,562	266,331	68,684	25.8%
PTLL-100	2,933,762	886,704	372,657	372,607	344,587	305,050	292,532	95.9%
PTLL-200	2,735,083	949,986	504,655	504,584	457,929	375,168	369,611	98.5%
PTLL-400	1,560,230	591,000	365,818	365,780	324,198	226,470	225,901	99.7%
PTLL-600	1,236,732	435,749	279,760	279,731	241,960	138,241	138,146	99.9%

1,094,874 raw MC events pass the full Z→μμ selection; the most (34%) come from PTLL-200. The AK4 jet step keeps 26% of inclusive (low-pT) Z events and ≥ 96% at pT(ℓℓ) > 100 GeV.

Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · "all" = events in the sample's LHE pT(ℓℓ) slice · 6

Yields by AK4 jet flavour (7.98 fb^{-1})

Channel	light	c	b	total
Z→ee	373,416.5	34,579.9	16,853.2	424,849.6
fraction	0.879	0.081	0.040	
Z→μμ	674,884.4	62,972.2	32,587.5	770,444.2
fraction	0.876	0.082	0.042	

20 files per sample (100 of 1887 DY files)

723,498 (Z→ee) and 1,094,874 (Z→μμ) raw events pass. The inclusive sample (pT($\ell\ell$) < 100 GeV) still carries large per-event weights, so the far tails (e.g. N_jet ≥ 6) rest on a few events. No b-tag requirement applied yet.

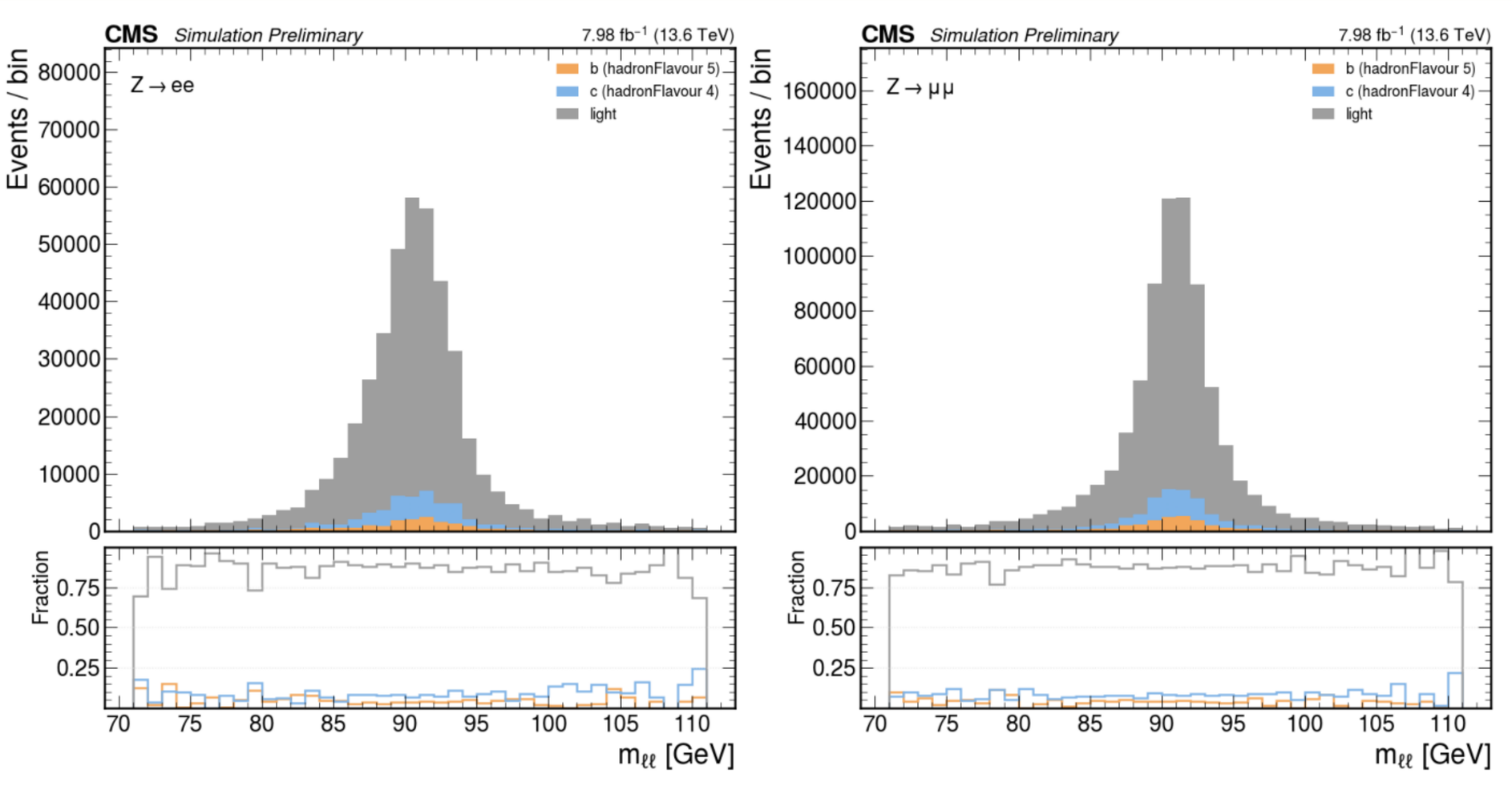
AK4: 1 file vs 20 files per sample

	1 file Z→ee	20 files Z→ee	1 file Z→μμ	20 files Z→μμ
Raw events selected	21,367	723,498	33,106	1,094,874
Scaled yield	403,007.9	424,849.6	784,702.2	770,444.2
b fraction	0.042	0.040	0.045	0.042
c fraction	0.106	0.081	0.083	0.082
light fraction	0.852	0.879	0.872	0.876

Same selection and code; 20× more files per sample, about 33× more raw events. The 1-file test had only N_eff ≈ 466 (Z→ee) / 915 (Z→μμ) effective events. The largest change is the Z→ee c fraction, 0.106 → 0.081, 1.7σ of the 1-file statistical error, so the 1-file numbers were a fair estimate. The AK8 vs AK4 comparison (1-file test only) is in the test deck.

Dilepton mass

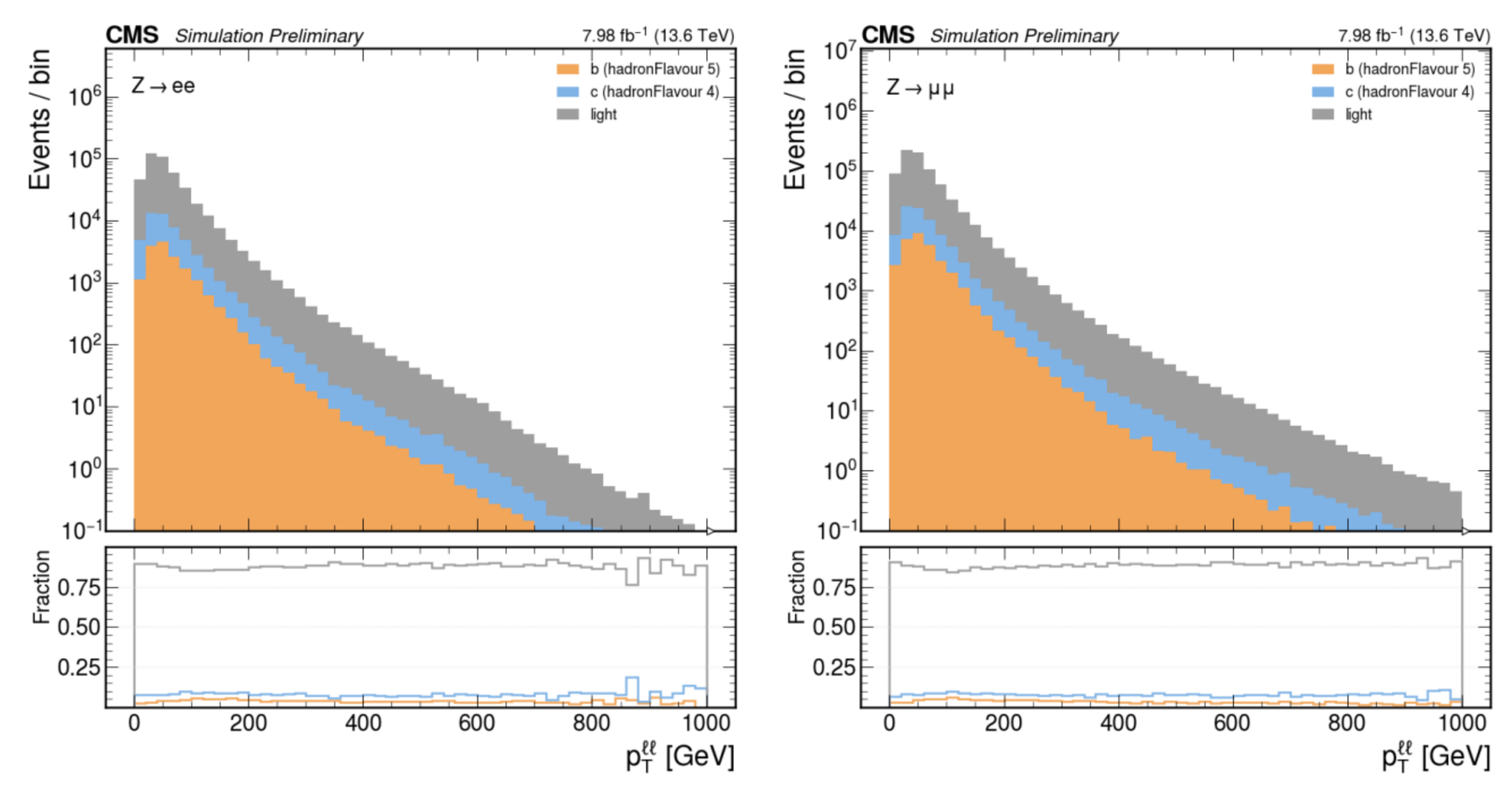
m(ℓℓ) after the full selection (71–111 GeV window): peaks at the Z mass in both channels. Linear y-axis.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 9

Dilepton pT

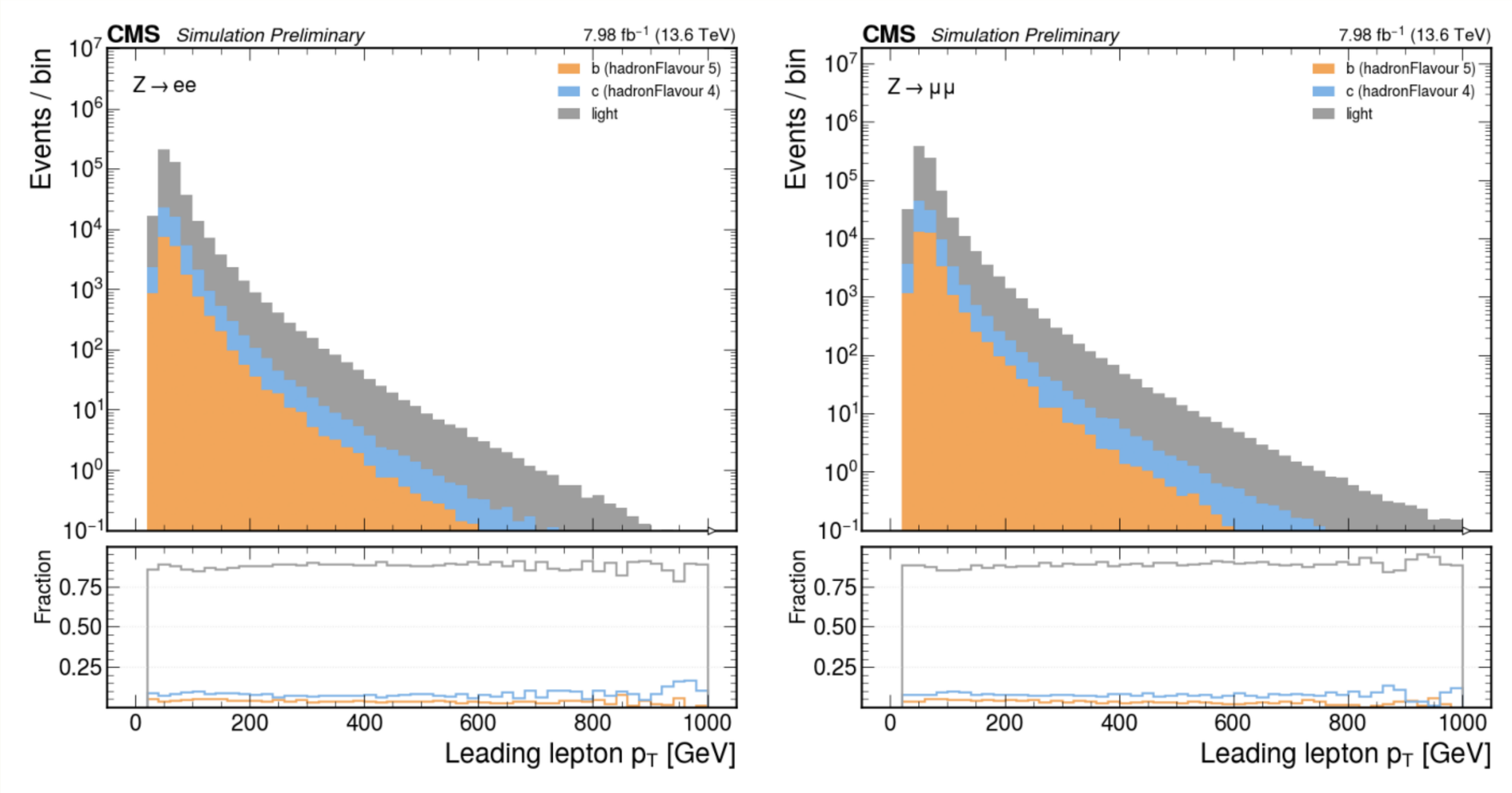
pT(ℓℓ) of the Z candidate after the full selection. No Z pT cut: dominated by low pT, unlike the boosted AK8 selection.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 10

Leading-lepton pT

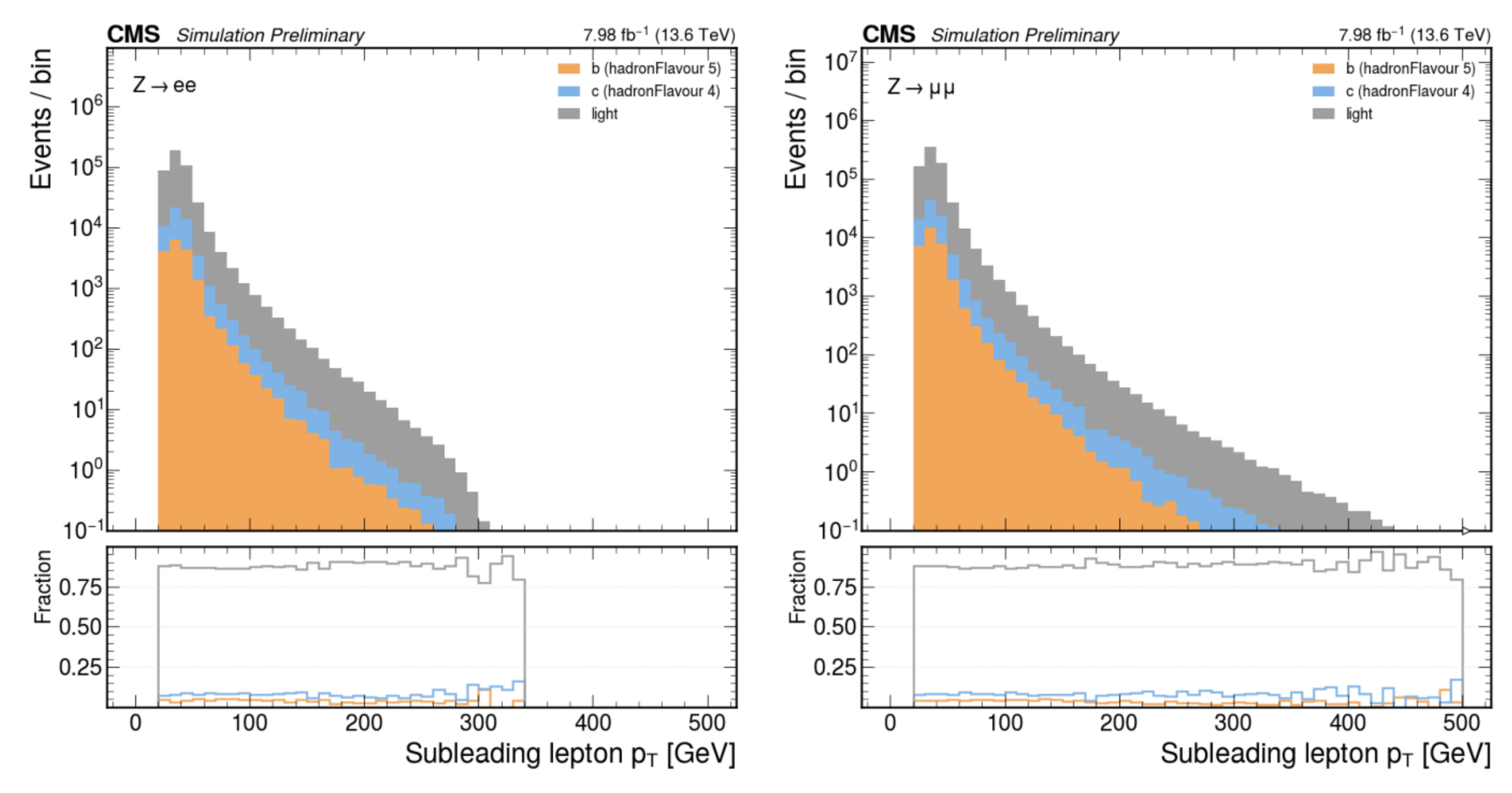
Leading lepton of the Z candidate, after the full selection (pT > 35 GeV required).



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 11

Subleading-lepton pT

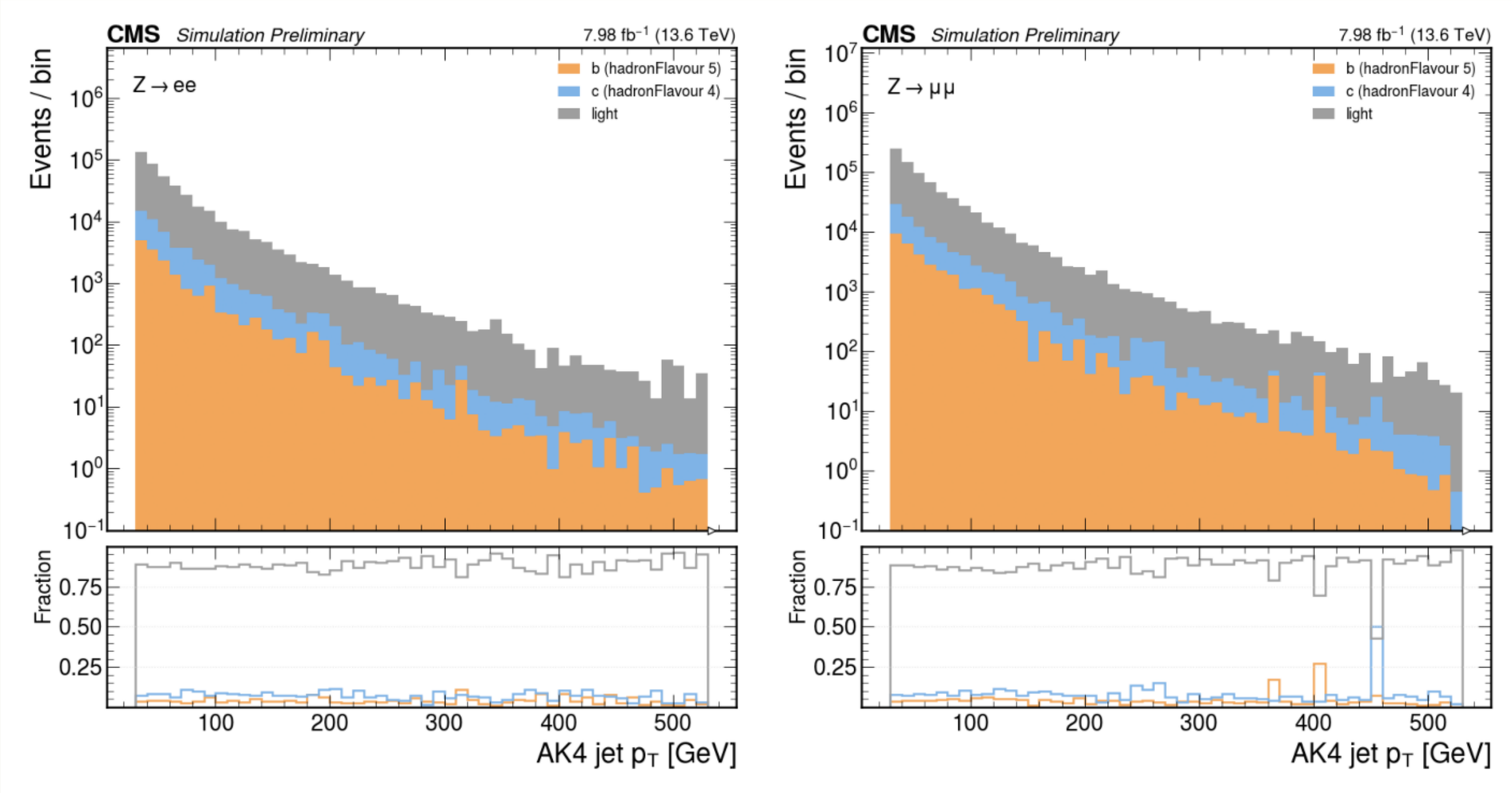
Subleading lepton of the Z candidate, after the full selection (pT > 25 GeV required).



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 12

Leading AK4 jet pT

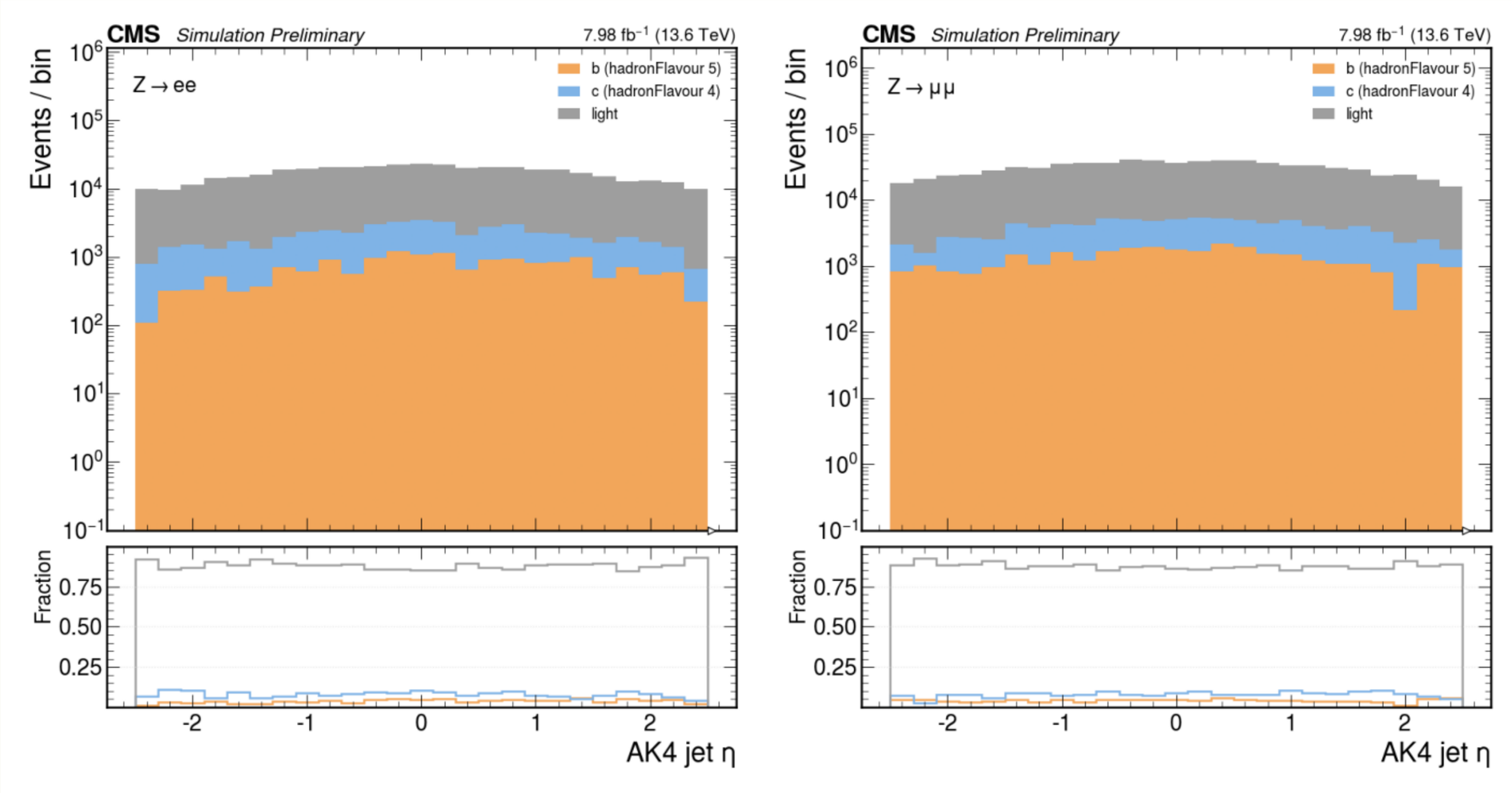
Leading lepton-cleaned AK4 jet with tight-lepton-veto ID (pT > 30 GeV required).



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 13

Leading AK4 jet η

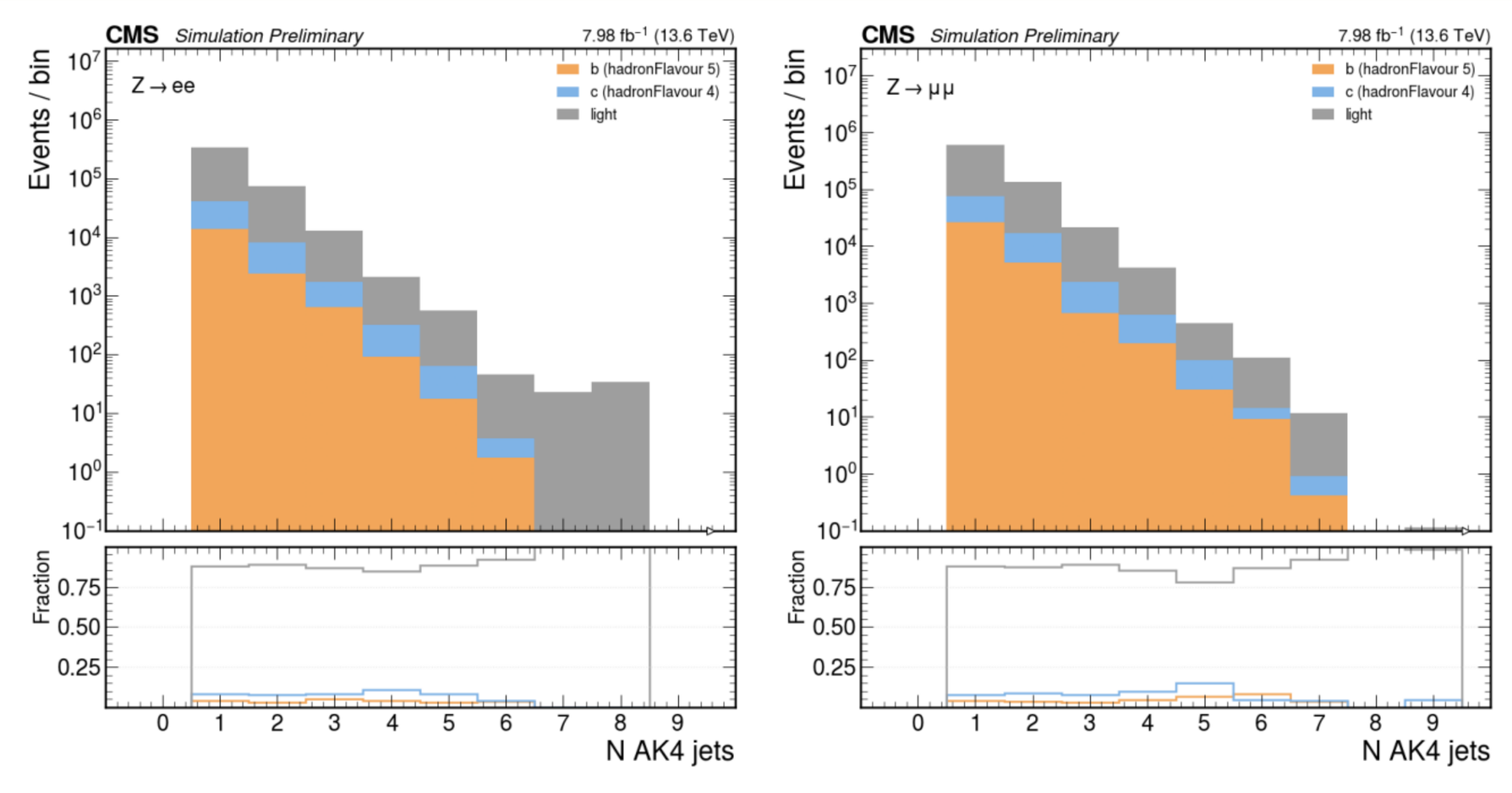
Pseudorapidity of the selected AK4 jet (|η| < 2.5 required).



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 14

AK4 jet multiplicity

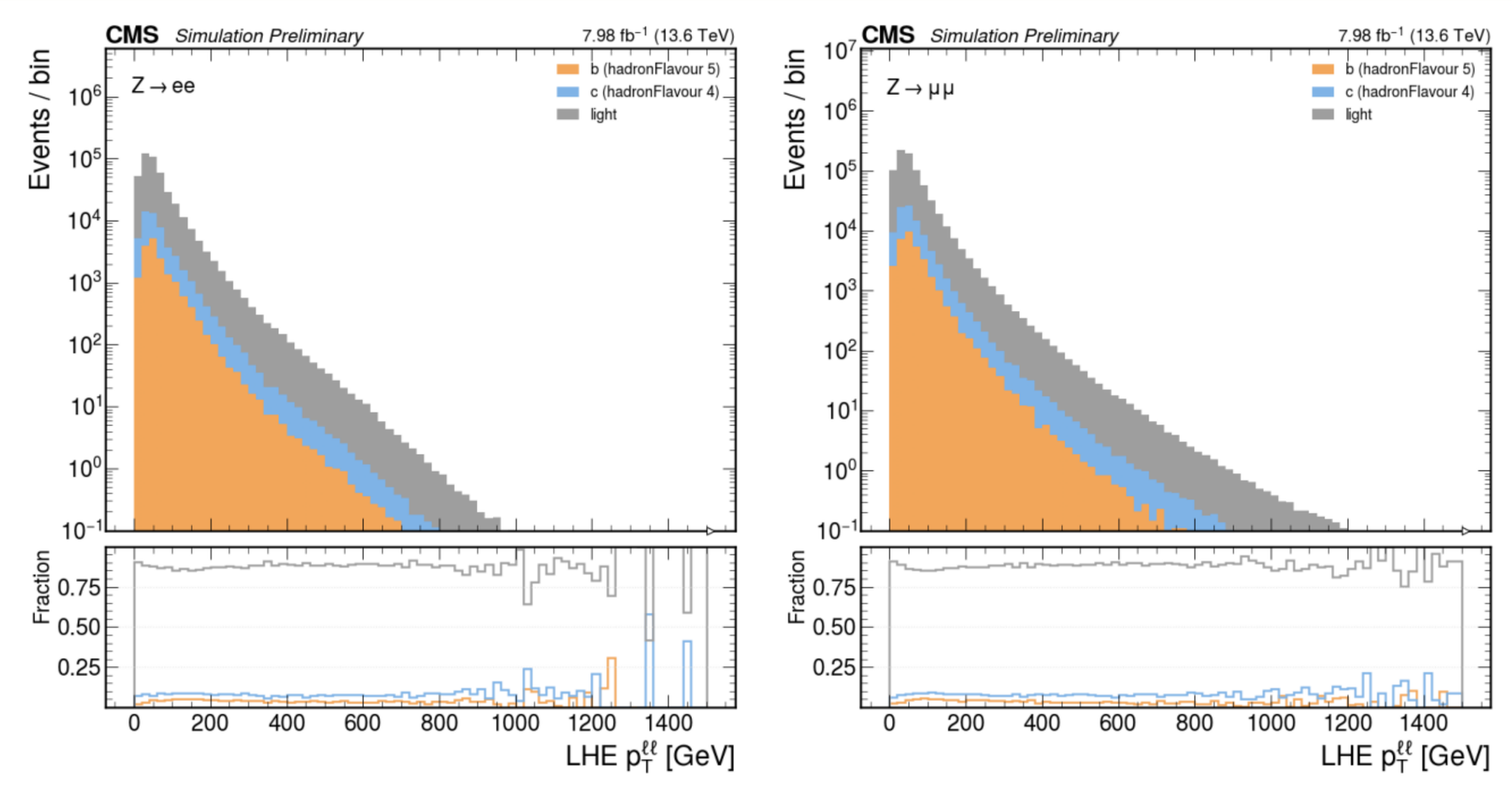
Number of selected AK4 jets (pT > 30 GeV, |η| < 2.5, ID, lepton-cleaned) in events passing the full selection.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 15

LHE pT($\ell\ell$)

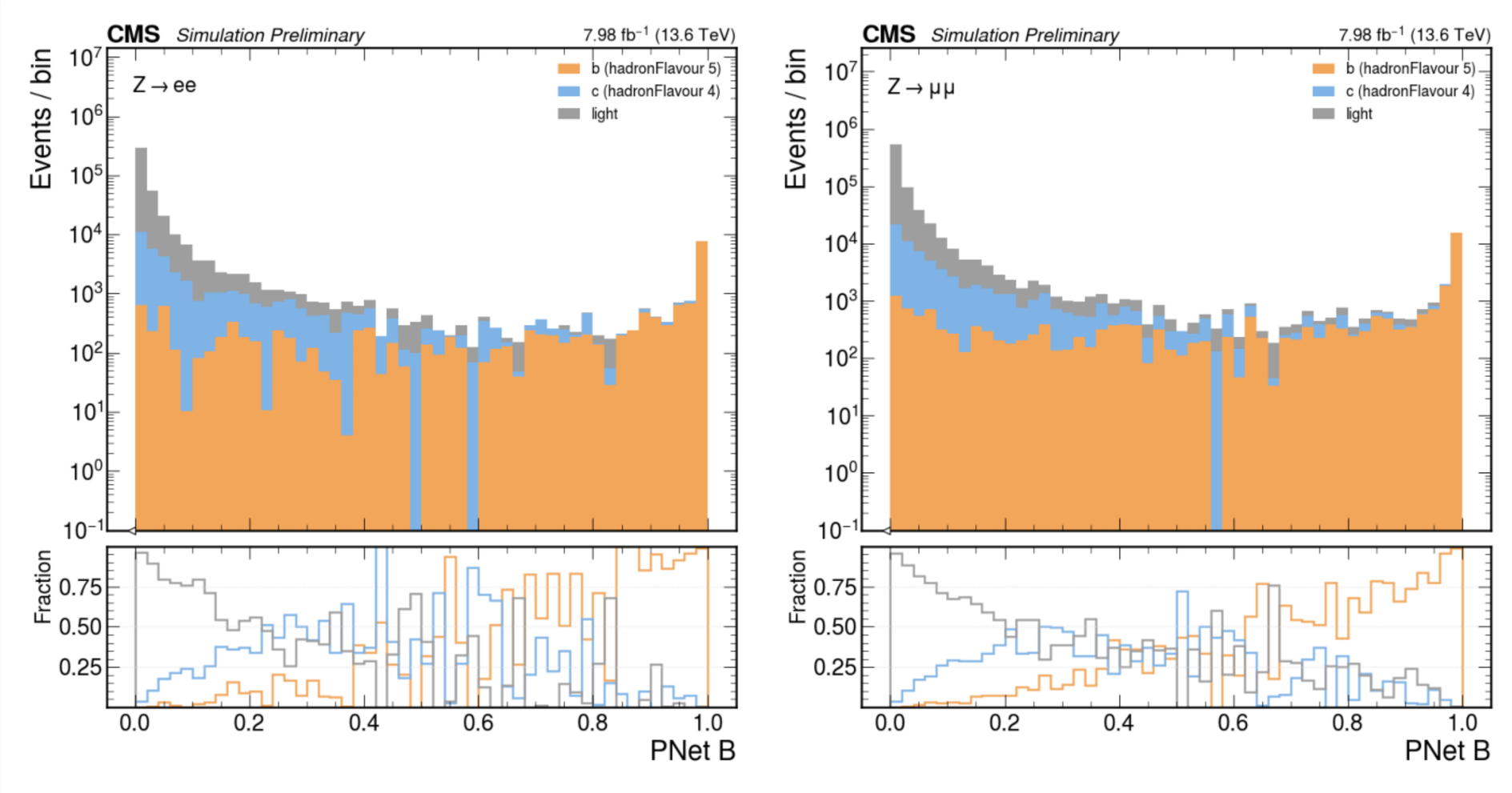
Generator-level pT($\ell\ell$) after the full selection: checks the sample stitching (slices at 100/200/400/600 GeV).



Summer22 DY→$\ell\ell$ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 16

Leading AK4 jet: PNet B

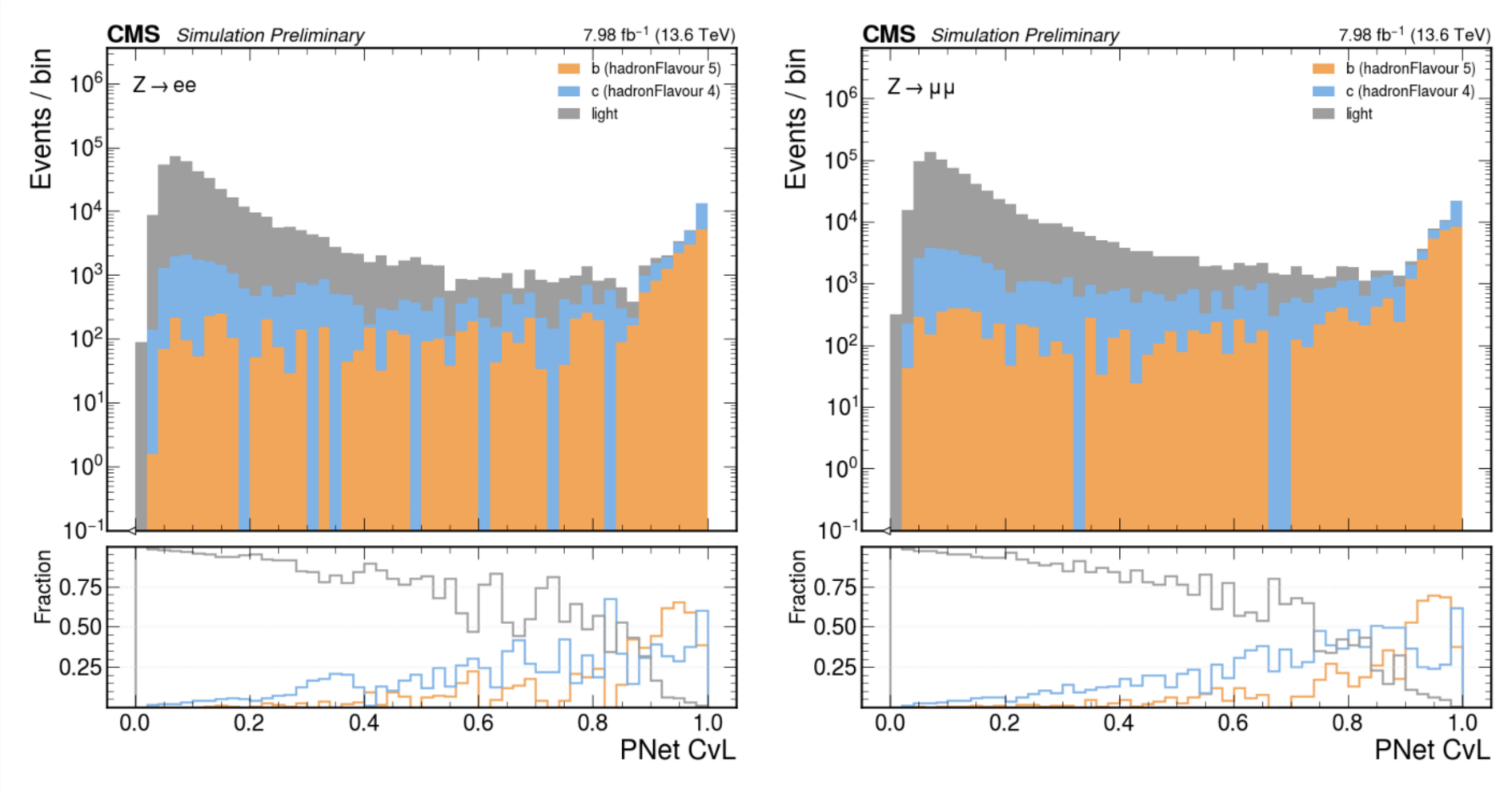
ParticleNet b-tag score of the selected jet; b jets pile up near 1, light jets near 0.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 17

Leading AK4 jet: PNet CvL

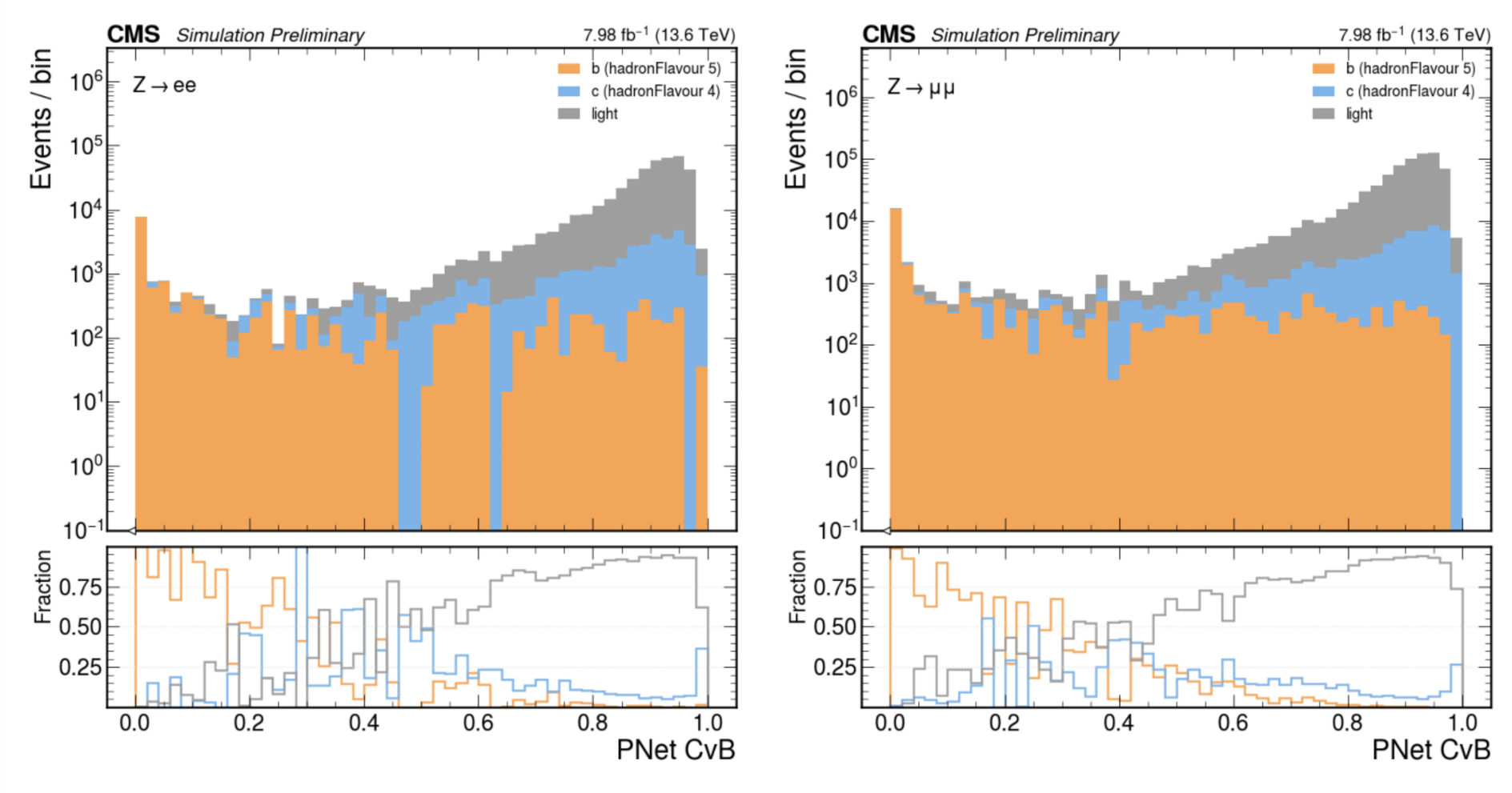
ParticleNet charm-vs-light score of the selected jet.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 18

Leading AK4 jet: PNet CvB

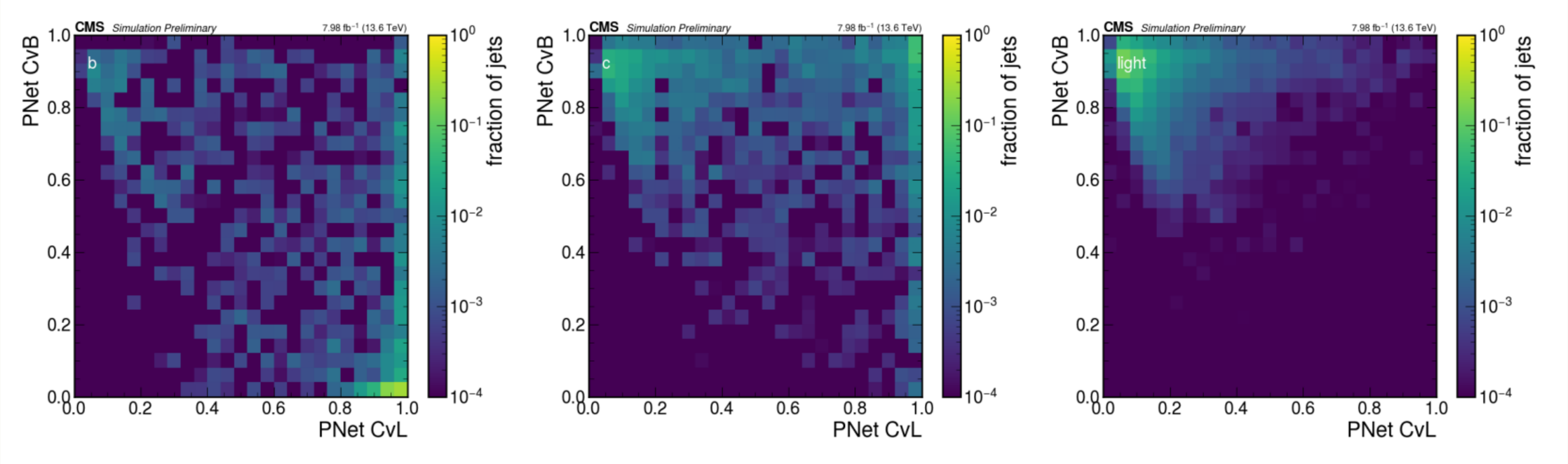
ParticleNet charm-vs-b score of the selected jet.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · stacked by leading-jet flavour; lower: fraction per bin · Z→ee left, Z→μμ right · 19

PNet charm-tagging plane: CvL vs CvB

Selected AK4 jets per flavour, both channels summed, normalized to unit area. b jets sit at high CvL and low CvB, light jets at low CvL and high CvB. c jets are at high CvB, but a large part of them overlaps the light-jet corner (low CvL), with the rest at high CvL.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · Z→ee + Z→μμ · 20

Jet flavour: what CMS recommends

CMS recommendation (BTV)

- Use the hadron-based flavour (hadronFlavour), the CMSSW default
- Generator B and C hadrons are clustered into the jet with ~zero momentum, so the jet itself is unchanged
- b if the jet has a B hadron, c if it has a C hadron but no B, light otherwise
- All BTV b/c-tag scale factors are given for this definition
- partonFlavour: mainly to separate gluon from uds jets

What we do

- Take Jet_hadronFlavour from NanoAOD for the leading jet: 5 = b, 4 = c, rest = light
- No matching of our own
- No b/bb or c/cc split: NanoAOD doesn't store hadron counts for AK4 jets
- partonFlavour only as a cross-check: agrees for 99% of b jets and 95% of c jets

Conclusion: we are on the right track

Our flavour labels are exactly the ones CMS recommends, so the BTV scale factors can be applied as they are. The c/b ratio in our MC (2.1 for Z→ee, 1.9 for Z→μμ) is a bit higher than, but in the same ballpark as, the CMS 13 TeV measurement, 1.62 ± 0.15 (PRD 102, 032007). Not a like-for-like comparison yet: ours is MC truth for the leading jet, without tagging.

Reference: CMS TWiki, Jet Flavour Identification (MC Truth)

https://twiki.cern.ch/twiki/bin/view/CMSPublic/SWGuideBTagMCTools#Hadron_parton_based_jet_flavour

hadronFlavour vs partonFlavour

hadronFlavour (default)

- Ghost-matched B / C hadrons clustered into the jet
- b (5) if any B hadron, else c (4) if any C hadron, else 0
- BTV standard; b/c-tag SFs are measured for it

partonFlavour

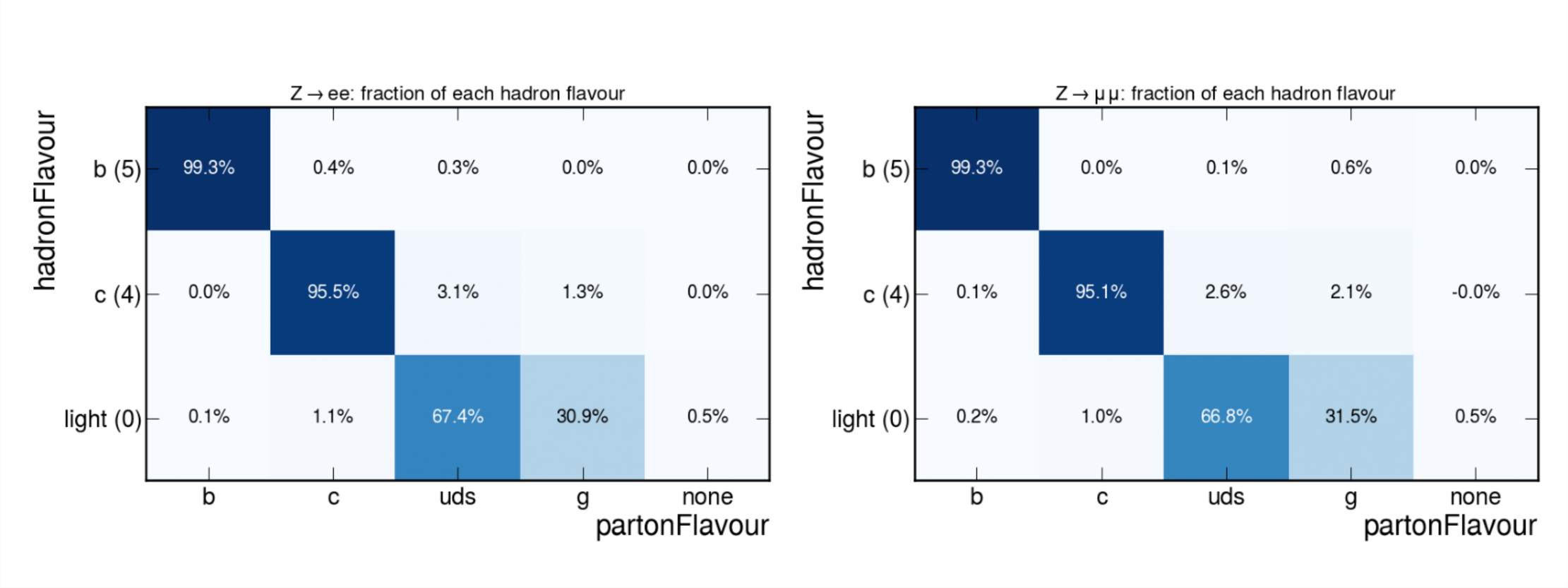
- Ghost-matched partons (last before hadronization) in the jet
- Hardest heavy parton wins: b (5), c (4), else hardest light parton
- Splits light jets into uds (1–3) and gluon (21); 0 = no parton matched

Fraction of scaled yield	hadron Z→ee	parton Z→ee	hadron Z→μμ	parton Z→μμ
b	0.040	0.041	0.042	0.044
c	0.081	0.087	0.082	0.087
light (hadron) / uds + g + none (parton)	0.879	0.872	0.876	0.870
of which g / none (parton)		0.273 / 0.004		0.278 / 0.004

Summer22 DY→$\ell\ell$ MC · 20 files per sample · AK4 jets · leading AK4 jet, scaled to 7.98 fb^{-1} · 22

Hadron vs parton flavour migration

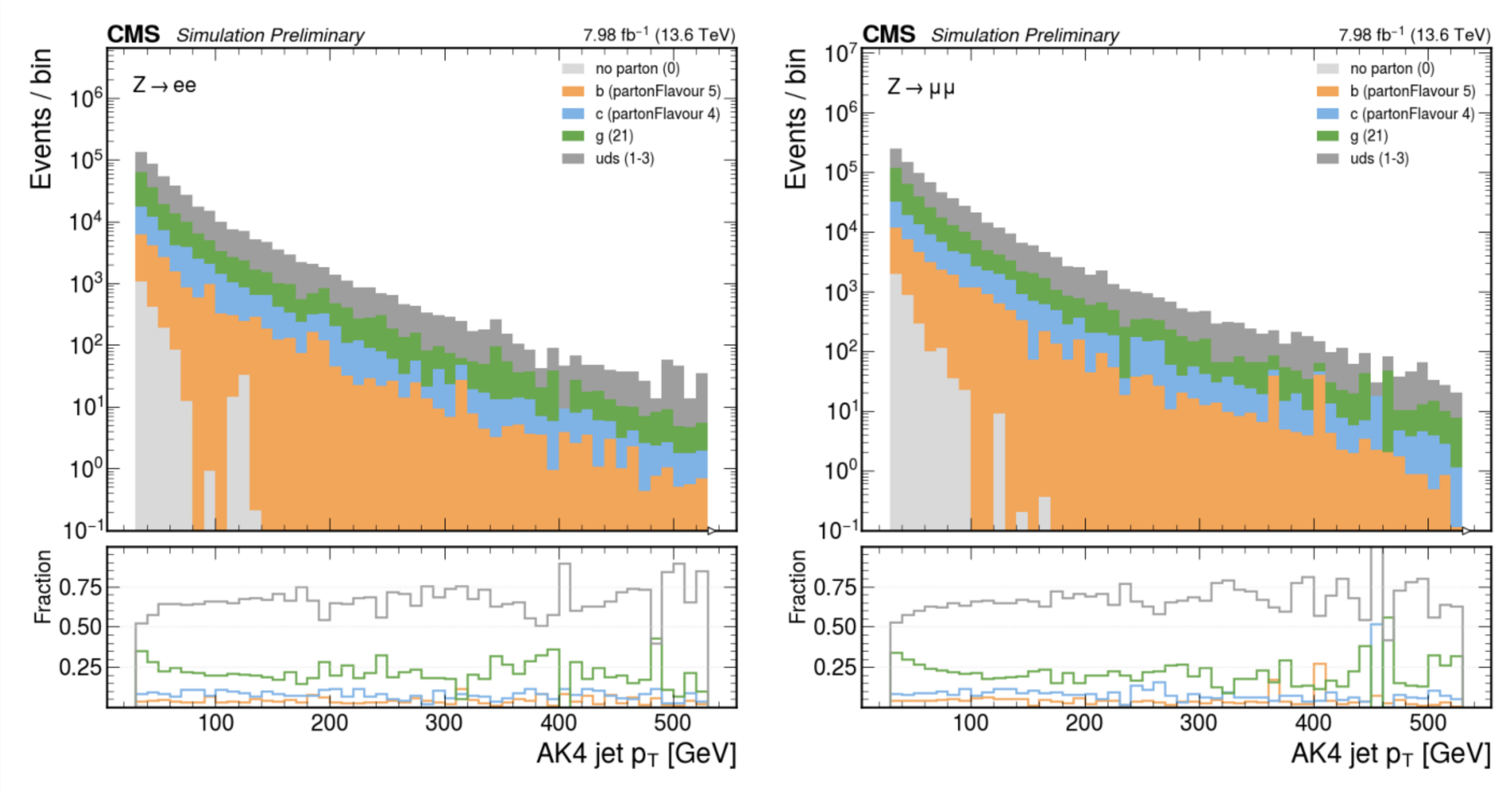
Each row shows where the jets of one hadronFlavour land in partonFlavour. 99% of hadron-b and 95% of hadron-c jets also have partonFlavour b / c (both channels). Hadron-light jets split into uds (67%) and gluon (31%); 1.2% carry a c or b parton without a matching hadron.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · rows: hadronFlavour, each sums to 100% (scaled) · small negative entries = NLO negative wei

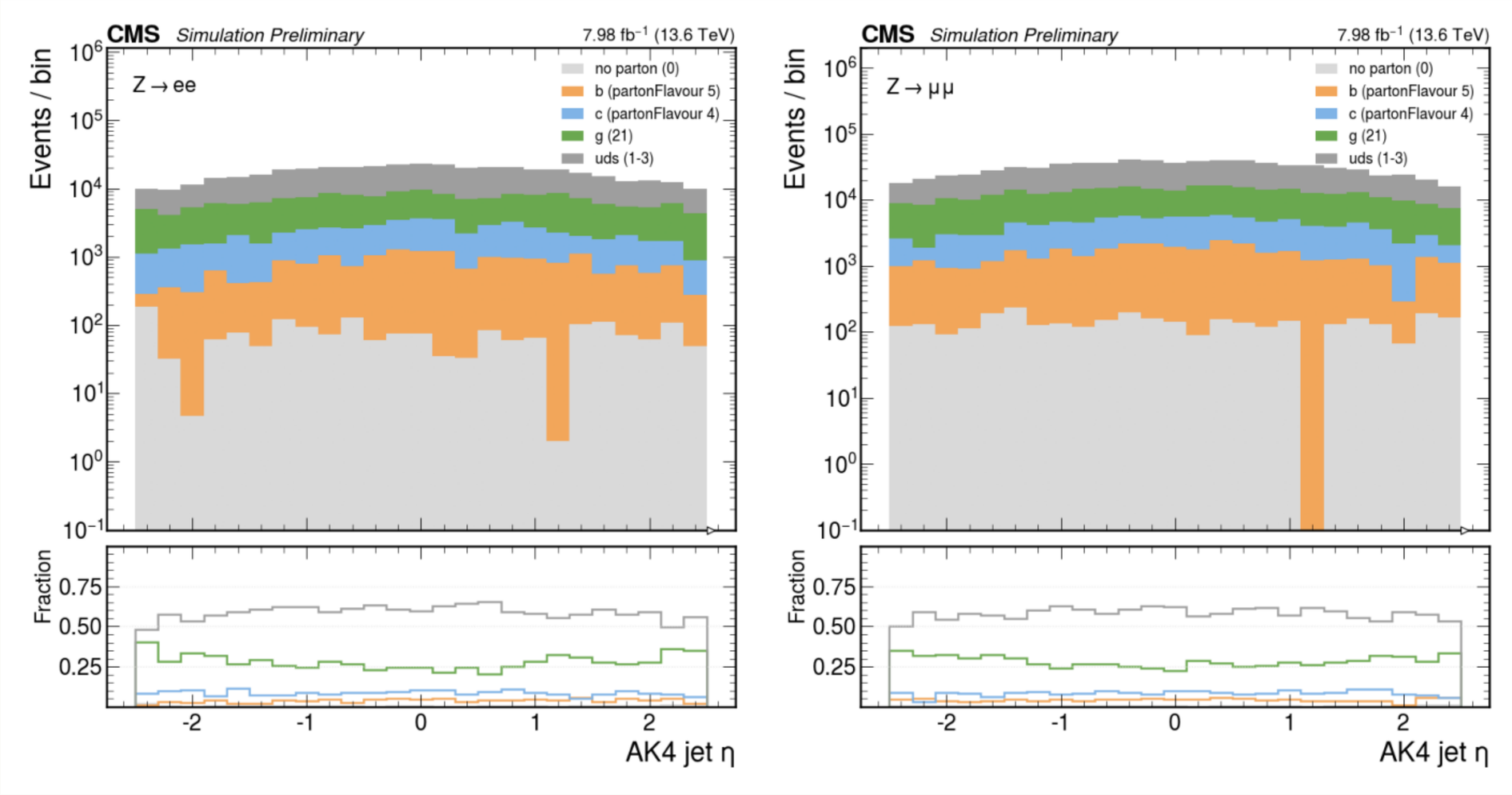
Leading AK4 jet pT, by partonFlavour

Same as page 13, stacked by |partonFlavour| of the leading jet.



Leading AK4 jet η, by partonFlavour

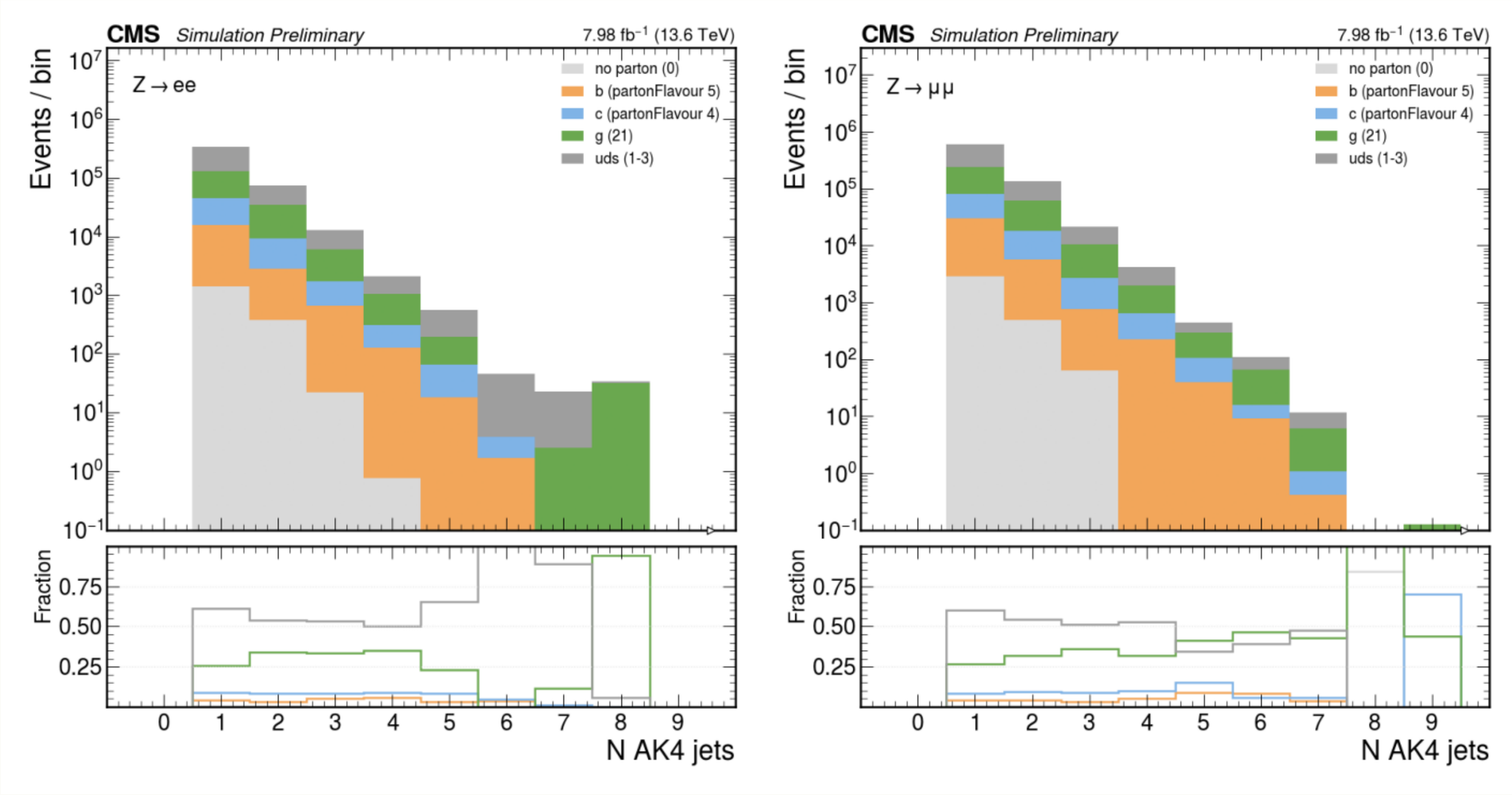
Same as page 14, stacked by |partonFlavour| of the leading jet.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · by leading-jet partonFlavour; lower: fraction per bin · Z→ee left, Z→μμ right · 25

AK4 jet multiplicity, by partonFlavour

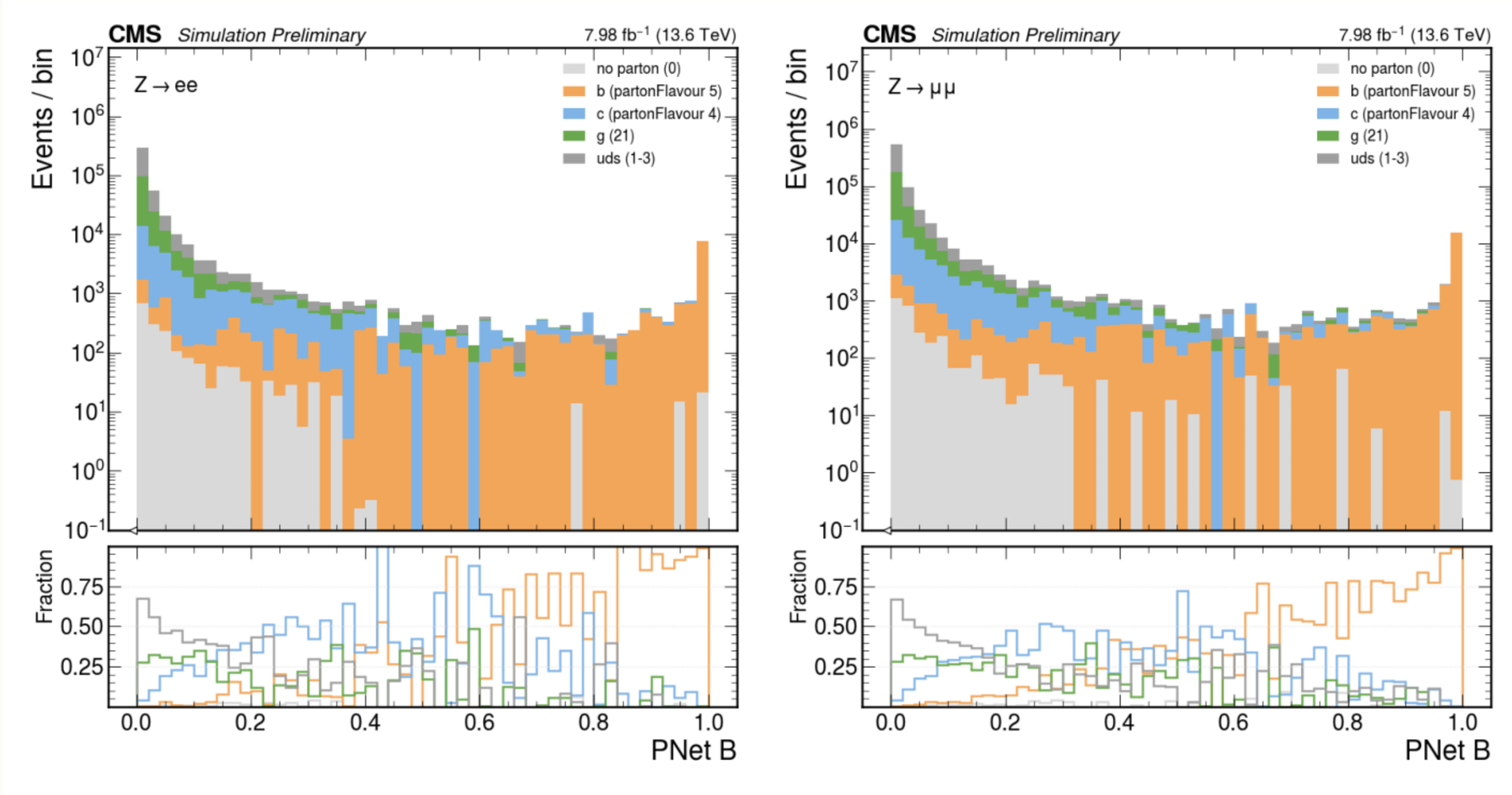
Same as page 15, stacked by |partonFlavour| of the leading jet.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · by leading-jet partonFlavour; lower: fraction per bin · Z→ee left, Z→μμ right · 26

PNet B, by partonFlavour

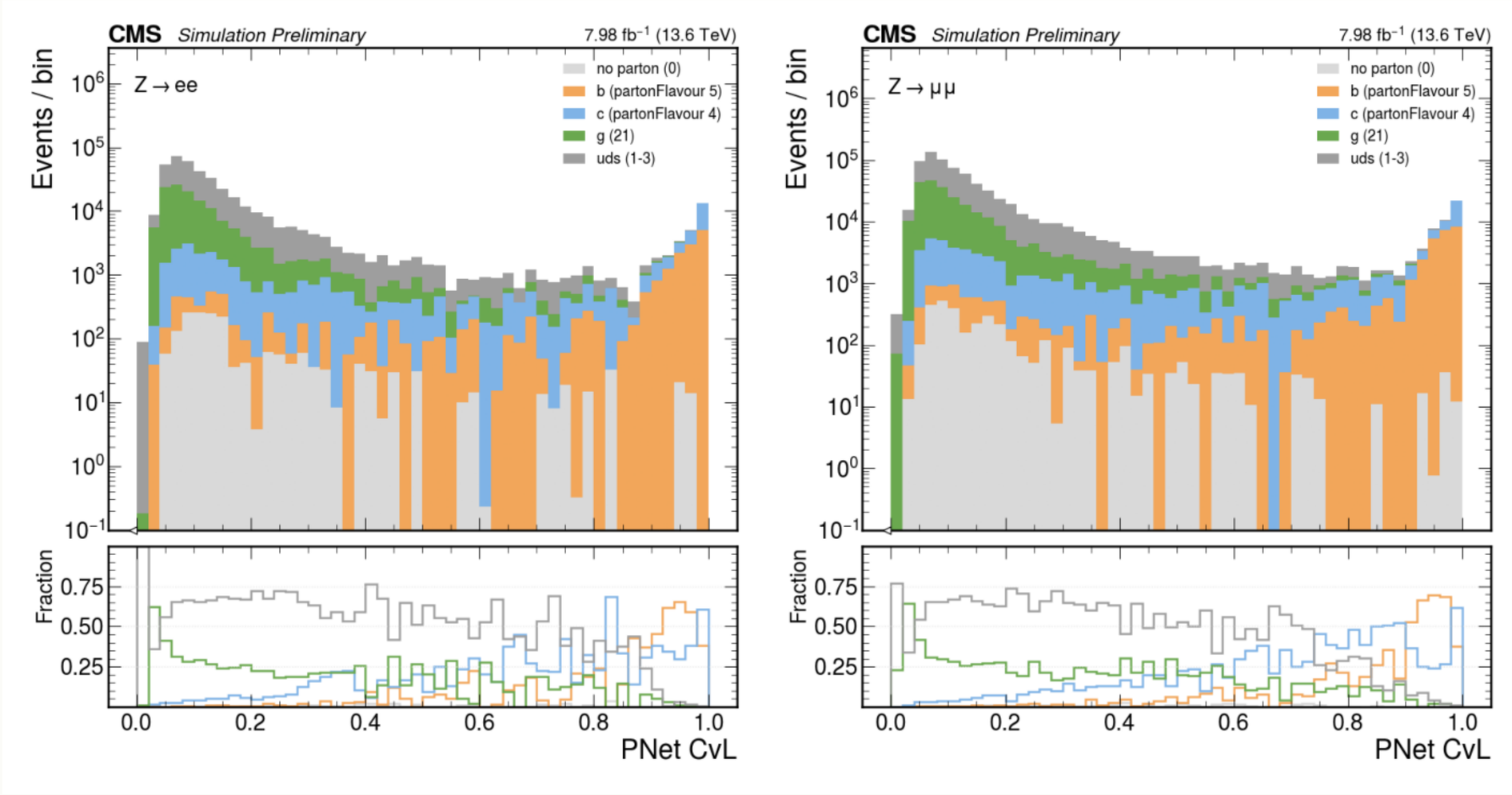
Same as page 17, stacked by |partonFlavour| of the leading jet.



Summer22 DY→ℓℓ MC · 20 files per sample · AK4 jets · by leading-jet partonFlavour; lower: fraction per bin · Z→ee left, Z→μμ right · 27

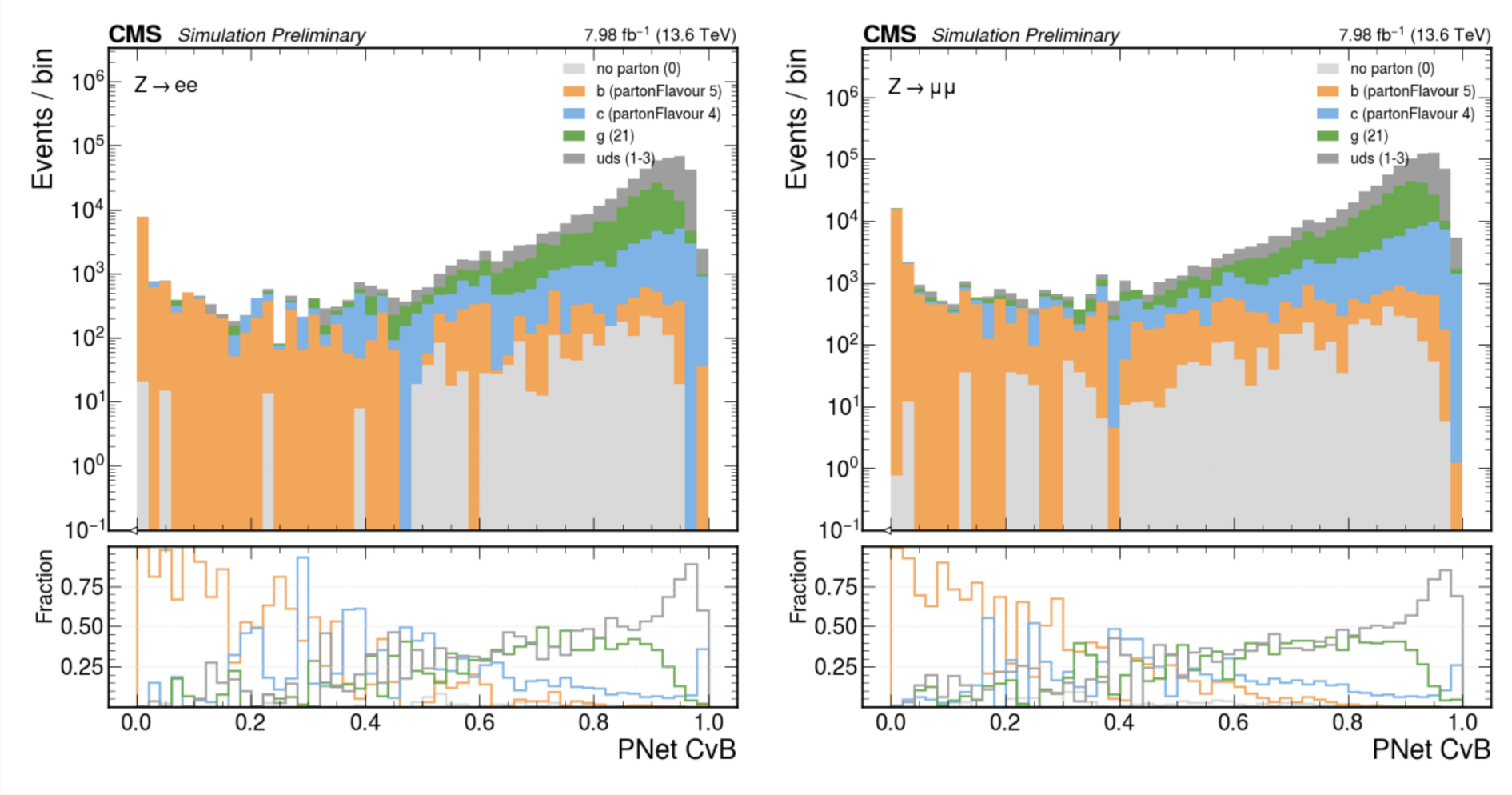
PNet CvL, by partonFlavour

Same as page 18, stacked by |partonFlavour| of the leading jet.



PNet CvB, by partonFlavour

Same as page 19, stacked by |partonFlavour| of the leading jet.



No-parton jets are pileup

What we saw

In 0.4% of events the leading jet has no parton attached (partonFlavour 0). They are all light jets, and they are soft: 83% are below 50 GeV, compared with 52% for all leading jets.

How we checked

A jet from the main collision should have a generator-level jet (GenJet) next to it. NanoAOD stores the match as Jet_genJetIdx, which is −1 when there is none within $\Delta R < 0.4$. 99% of the no-parton jets have no GenJet, against 0.02% for b, c and uds jets. No generator partner means the jet comes from another collision in the same bunch crossing: pileup.

Why we keep them

We can't use generator information in data, so we can't remove these jets by truth in MC either. Data has pileup jets too, and the MC has to describe them. Since they are light jets and a tiny fraction of the yield, Z+b and Z+c are not affected.

Bottom line: expected, small, and harmless for the b/c ratio

No action needed. Our AK4 jets are PUPPI jets, which already remove most pileup. If we ever want fewer, the handle is a reco-level cut (e.g. a higher jet pT threshold), applied the same way to data and MC.

Leading jets with partonFlavour 0 are pileup

A jet with no GenJet (pT > 10 GeV) within ΔR < 0.4 (Jet_genJetIdx = −1) is a pileup jet. 395 of 399 (99%) leading jets with partonFlavour 0 have no GenJet match (131 Z→ee, 264 Z→μμ).

Leading-jet partonFlavour	Z→ee no GenJet / all	Z→ee %	Z→μμ no GenJet / all	Z→μμ %
b	5 / 33,829	0.01	6 / 50,225	0.01
c	14 / 63,633	0.02	26 / 95,070	0.03
uds	97 / 451,608	0.02	187 / 684,307	0.03
g	363 / 174,296	0.21	621 / 265,005	0.23
none	131 / 132	99.24	264 / 267	98.88
all	610 / 723,498	0.08	1,104 / 1,094,874	0.10

- Not every pileup jet has partonFlavour 0: most unmatched leading jets carry a gluon label (a soft ISR/UE gluon is ghost-matched, but there is no GenJet).
- Pileup barely reaches the heavy-flavour yields: 7 of 82,303 hadron-b and 36 of 151,037 hadron-c leading jets are unmatched. Scaled, 1.8% (Z→ee) and 1.7% (Z→μμ) of leading jets are unmatched. This is above the raw fraction because the unmatched jets sit in the high-weight inclusive sample, but it is close to the 1-file value, so it is not a statistical fluke.

N_jet bumps go away with more MC

20 files per sample instead of 1. Z→ee N_jet = 6: 45.5 (N_eff 0.7) → 45.5 (N_eff 0.7); Z→μμ N_jet = 6: 108.4 (N_eff 2.9) → 108.4 (N_eff 2.9), N_jet = 8: 0.0 (N_eff 0.0) → 0.0 (N_eff 0.0). Both spectra now fall smoothly up to N_jet = 7. The new Z→ee N_jet = 8 point, 33.7 (N_eff 1.1), is again a single inclusive-sample event: N_jet ≥ 6 still needs the full run.

